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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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Java-Based Investment Portfolio Analysis, Optimization and Recommendation Framework Using Bayesian Networks

CSA0911 Programming in JAVA

Guided by
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PROBLEM STATEMENT

- Investors track blue-chip stocks, real estate, and precious metals across disconnected platforms, with no single system unifying these asset classes or showing a clear market signal.
- Existing tools display raw prices and values but lack probabilistic risk analysis, leaving users without a data-driven Bullish/Bearish/Neutral indication to guide decisions.
- There is no lightweight, Java-based solution tailored to Indian market data — NSE blue-chips, Chennai real estate, and gold/silver — that combines tracking with predictive insight.

PROJECT SCOPE & OBJECTIVES

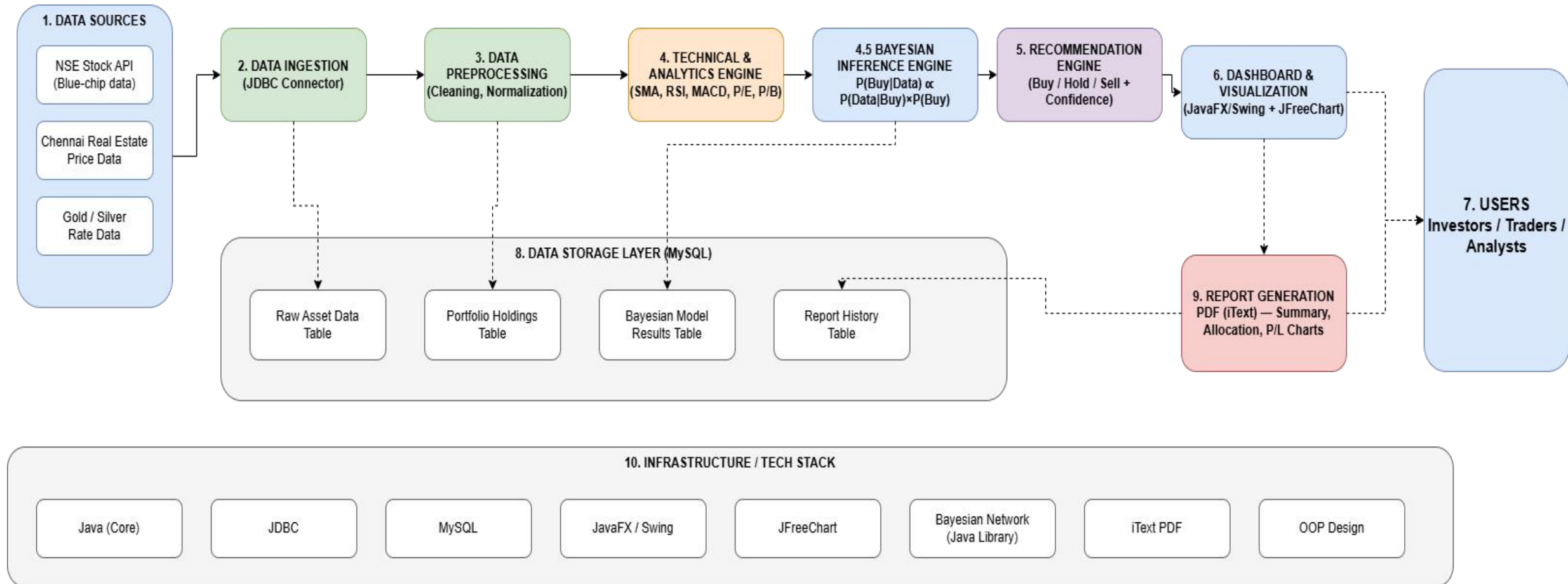
- Design a unified portfolio management system for stocks, real estate, and physical assets with manual entry, real-time value tracking, and risk scoring.
- Build a Bayesian recommendation engine that classifies each asset as Bullish, Bearish, or Neutral, and outputs a Buy/Hold/Sell call with a confidence score.
- Deliver an interactive dashboard with technical charts and downloadable PDF reports for portfolio summary, allocation, and profit/loss.



Engineer to Excel



SYSTEM ARCHITECTURE



LITERATURE REVIEW

S.No	Research Paper	Author(s)	Advantages	Disadvantages	Technologies Used
1	From Classical Optimization to Bayesian Integration: A Comprehensive Analysis of Systematic Portfolio Management (2026)	Verma, A.K. & Barkam, S.	Compares Bayesian, mean-variance, and Black-Litterman models on real stock data; shows Bayesian integration improves allocation stability	Limited to 10 U.S. large-cap stocks; no real estate or gold asset class covered	Bayesian Modelling, Black-Litterman, Monte Carlo Simulation
2	A Comparative Study of Bayesian Portfolio Optimization: Evidence from U.S. Large-Cap AI-Related Stocks (2026)	Mikhailm, M.	Uses posterior sampling (2000 draws) to outperform Markowitz benchmark; strong empirical validation over 3-year training/1-year test split	High computational cost of posterior sampling; sector-specific	Bayesian Inference, Markowitz MVO, TradingView/Tiingo Data
3	Advanced Investing with Deep Learning for Risk-Aligned Portfolio Optimization (2025)	(PMC/NCBI published)	Combines LSTM forecasts with optimization strategies for better risk-adjusted returns; validated on 2023–2024 real market data	Deep learning models are compute-heavy and less interpretable than Bayesian probability output	LSTM, 1D-CNN, Mean-Variance Optimization
4	A Framework for Gold Price Prediction Combining Classical and Intelligent Methods with Financial, Economic, and Sentiment Data Fusion (2025)	(MDPI journal)	Fuses econometric, ML, and sentiment data for gold forecasting; validated on 11-year dataset across market conditions	Complex multi-model fusion increases system overhead for a lightweight app	Machine Learning, Sentiment Analysis, Econometric Modelling
5	Stock Recommendations for Individual Investors: A Temporal Graph Network Approach with Mean-Variance Efficient Sampling (2024)	Sanz-Cruzado, J. et al.	Tailors recommendations to individual investor behavior rather than generic market signals; introduces FAR-Trans dataset for finance recommendation research	Graph-based models need large transaction datasets; not designed for real-time dashboards	Temporal Graph Networks, Mean-Variance Sampling

Module 1 – Financial Data Collection and Portfolio Database Management

- Collect and organize financial data for Indian stocks, US stocks, mutual funds, ETFs, and real estate, including asset details, transactions, purchase price, quantity, and current value.
- Store and maintain five years of historical financial data for supported assets, creating a structured database for portfolio analysis and investment evaluation.
- Allow users to add, update, delete, and retrieve portfolio records, while maintaining accurate and organized information for each individual asset.
- Provide reliable financial and historical market data to the analysis, Bayesian processing, visualization, and reporting modules.

Module 2 – Integrated Fundamental and Technical Investment Analysis Engine

- Perform fundamental analysis using financial parameters such as revenue, profit, earnings, valuation ratios, and other relevant indicators to evaluate the financial strength of supported assets.
- Perform technical analysis using historical price and volume data with indicators such as moving averages, RSI, MACD, and other technical measures to identify market trends and patterns.
- Combine fundamental and technical analysis results to provide a comprehensive evaluation of individual assets and their potential investment performance.
- Generate structured analytical results from fundamental metrics, technical indicators, price trends, and historical data for use in Bayesian analysis and portfolio decision-making.

Module 3 – Bayesian Investment Probability and Decision Analysis Engine

- Apply Bayesian analysis to historical financial data and analytical results to estimate the probability of different investment outcomes for supported assets.
- Use historical returns, volatility, fundamental metrics, technical indicators, and other relevant parameters as evidence for calculating probability-based investment results.
- Generate investment indications such as Buy, Hold, or Sell based on the calculated Bayesian probabilities and available financial evidence.
- Present Bayesian probability results along with the underlying fundamental, technical, and historical information to provide transparent investment decision support.

Module 4 – Interactive Financial Visualization and Portfolio User Interface

- Provide an interactive user interface for viewing portfolio holdings, investment value, returns, profit or loss, asset allocation, and analysis results in a structured manner.
- Display financial information through interactive charts, graphs, tables, and performance indicators to simplify the interpretation of complex investment data.
- Visualize historical trends, fundamental metrics, technical indicators, portfolio performance, and Bayesian results for effective comparison across different assets.
- Provide downloadable financial reports combining portfolio information, analytical results, visualizations, and Bayesian insights into a consolidated investment report.

EVALUATION METRICS

- Technical Indicators — SMA 50, SMA 200, RSI, and MACD are calculated for each asset and checked against actual price charts for correctness.
- Bayesian Formula Accuracy — the posterior probability calculation ($P(\text{Buy}|\text{Data}) \propto P(\text{Data}|\text{Buy}) \times P(\text{Buy})$) is validated by comparing predicted sentiment against real market movement.
- Volatility Accuracy — standard deviation of daily returns (annualized volatility %) is compared against actual historical price fluctuation for each asset.
- Fundamental Ratios — P/E Ratio, P/B Ratio, and ROI are cross-checked against real company financials to confirm the accuracy of displayed valuation metrics.

EXPECTED OUTCOMES

- A working dashboard showing Total Value and overall Risk Score at the top, giving users an instant snapshot of their entire portfolio.
- A Bayesian recommendation card per asset showing a Buy/Sell/Hold call with confidence percentage, current gain, volatility, and market sentiment.
- A candlestick Technical Analysis Chart with volume bars for any selected asset, letting users visually track price trends over time.
- A downloadable PDF report covering portfolio summary, allocation chart, holdings table, and profit/loss chart for offline review.

CONCLUSION

- The project bridges the gap between passive asset tracking and active, data-driven decision-making by unifying stocks, real estate, and metals into one dashboard.
- Bayesian modelling adds a probabilistic edge over conventional tracking tools, converting raw price and volatility data into a confidence-scored Buy/Hold/Sell signal.
- The modular architecture (Data, Analytics, Recommendation, Reporting) ensures scalability for future asset classes, allowing sources like mutual funds or crypto to be added later.
- This system empowers everyday investors in India with an accessible, intelligent portfolio tool, reflecting practical use of databases and probabilistic modelling learned through the course.

THANK YOU

PROOF

